

Project Introduction: Asset Allocation and Portfolio Construction

In this project, we will construct a **strategic investment portfolio** using historical market data. The portfolio is built from a selection of **diversified asset classes** to balance risk and return across different economic sectors and geographies.

We use historical price data from **Yahoo Finance** for the following ETFs and indices:

- **SPY** – U.S. Equities (S&P 500 Index)
- **QQQ** – Nasdaq-100 (focused on Growth Stocks / Technology-sector)
- **^GDAXI** – German Stocks (DAX Index)
- **VNQ** – Real Estate Investment Trusts (REITs)
- **TLT** – Long-term U.S. Government Bonds
- **GLD** – Gold

Objective

Our goal is to identify an **optimal portfolio allocation** based on modern portfolio theory. We analyze the portfolio using **statistical and econometric tools** like statistical testing, OLS-regression and optimization.

Methodological Steps

1. **Data Processing & Jarque-Bera Normality Test**
 - Fetch and clean asset price data
 - Calculate log-returns
 - Test for normality of returns using the Jarque-Bera test
2. **Investment Opportunity Set & Efficient Frontier**
 - Estimate expected returns and covariance matrix
 - Construct the mean-variance efficient frontier
 - Visualize individual asset risk-return profiles
3. **Tangency Portfolio vs. Optimal Complete Portfolio**
 - Calculation of the tangency portfolio that maximizes the Sharpe Ratio
 - Determine the investor-specific optimal allocation using risk aversion
4. **Risk Decomposition via OLS Regression**
 - Decompose portfolio return into:
 - **Systematic Return (β)**
 - **Idiosyncratic Return (α)**
 - Analyze corresponding risk components
5. **Expected vs. Realized Performance**
 - Evaluate performance on test data
 - Compare expected return/volatility with realized out-of-sample results

Our project offers a complete, data-driven exploration of **modern portfolio construction**.

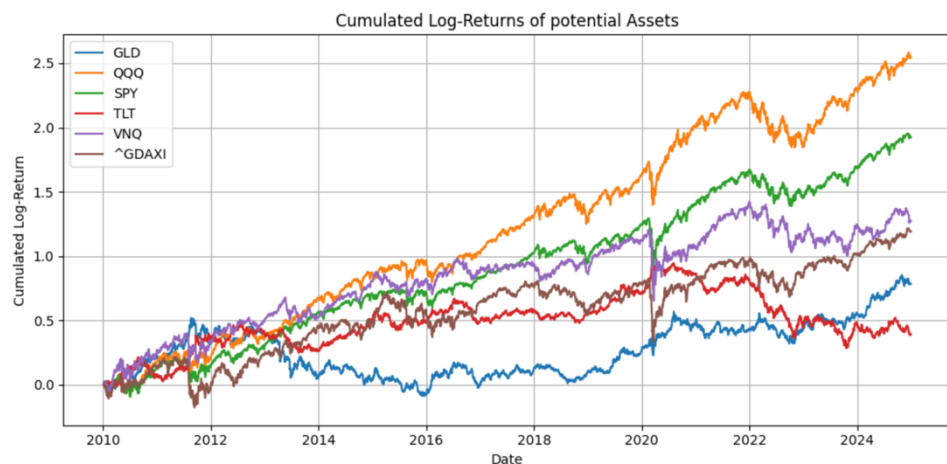
Selected Assets

Annualized Returns:

GLD	0.065557
QQQ	0.194135
SPY	0.145440
TLT	0.038552
VNQ	0.108291

Annualized Volatility:

GLD	0.156109
QQQ	0.205678
SPY	0.171260
TLT	0.154054
VNQ	0.207914



1) Testing for Normal Distribution: Jarque-Bera Test

The **Jarque-Bera test** checks if our data follows a **normal distribution** based on its **skewness** and **kurtosis**.

In finance, especially when using models like the **Capital Asset Pricing Model (CAPM)**, normality of returns is often assumed. This is important because many statistical methods (e.g. linear regression) are based on the idea that residuals or input variables behave "nicely" — meaning they're symmetrically distributed without extreme outliers.

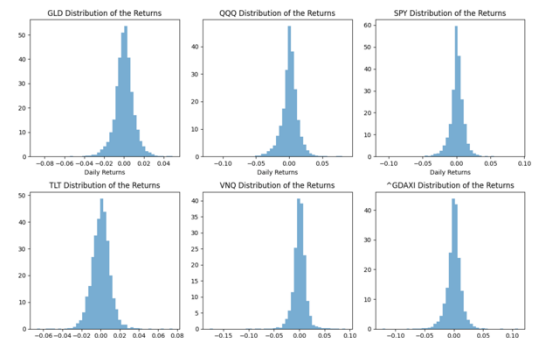
The Jarque-Bera test statistic is:

$$JB = \frac{n}{6} \left(\text{Skewness}^2 + \frac{1}{4}(\text{Kurtosis} - 3)^2 \right)$$

- **Skewness** tells us if the distribution is symmetric or lopsided
- **Kurtosis** tells us how heavy the tails are (i.e. how likely extreme values are)

If the resulting **p-value is small (< 0.05)**, we reject the assumption of normality — which can be not good when we apply CAPM to real data.

	JB statistic	p-value	Skewness	Kurtosis	Normal Distr.
GLD	3033.90805	0.0	-0.399641	7.359288	Nein
QQQ	5582.334666	0.0	-0.356784	8.96927	Nein
SPY	17837.234949	0.0	-0.524936	13.694871	Nein
TLT	1840.557209	0.0	0.057182	6.450091	Nein
VNQ	38779.735133	0.0	-0.754637	18.77312	Nein
GDAXI	7903.236202	0.0	-0.311418	10.125974	Nein



Why our Asset Returns Are Not Normally Distributed

In theory, we want our financial returns to follow a normal distribution (clean, symmetric, and with strange outliers). But our example data, return distributions are usually not gaussian:

- They have **fat tails**, meaning extreme losses or gains happen more frequently than a normal model would estimate.
- Returns can be **skewed**, especially during crises or bubbles when the stock-market goes asymmetrically.
- Real events — shocks (z.B. Corona), panic selling, **black swans** — are **extreme events** that disprove the assumptions of normality.

That's why our assets fail the test. That doesn't mean we can't use CAPM — but it reminds us to consider model limitations when interpreting results.

Correlation Analysis in Portfolio Allocation

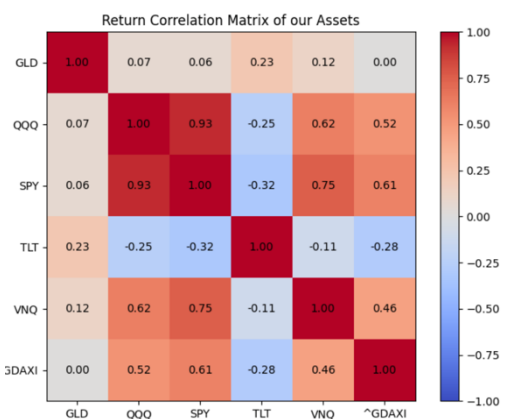
Correlation analysis helps to us understand how different asset classes move relative to each other. This is key for **diversification**

The correlation between two return variables R_i and R_j is defined as:

$$\rho_{ij} = \frac{\text{Cov}(R_i, R_j)}{\sigma_i \cdot \sigma_j}$$

Where:

- $\rho_{ij} \in [-1, 1]$: 1 = perfectly positive, -1 = perfectly negative correlation
- $\text{Cov}(R_i, R_j)$: covariance of returns
- σ_i, σ_j : standard deviations of returns



Interpretation:

- **SPY** (S&P 500) and **QQQ** (Nasdaq 100) are **highly correlated** ($\rho = 0.93$) due to strong overlap in large-cap tech stocks.
- **TLT** (long-term U.S. bonds) shows **negative correlation** with equities ($\rho = -0.32$), acting as a defensive hedge.
- **GLD** (Gold) is **almost uncorrelated** with both stocks and bonds ($\rho \approx 0$), making it a strong diversifier.
- **GDAXI** (German Stocks) is **moderately correlated** with U.S. equities, reflecting global economic connection.
- **VNQ** (REITs) shows **moderate correlation** with stocks due to their equity-like returns.

Conclusion:

Diversifying assets with **low or negative correlation** improves the **risk-return trade-off** of the portfolio by reducing overall volatility.

2) Investment Opportunity Set and Efficient Frontier

Characterizing the Efficient Frontier and Tangency Portfolio

The **investment opportunity set** consists of all possible portfolios formed by different combinations of risky assets. Within this set, the **Efficient Frontier** identifies the subset of portfolios that offer the **maximum expected return for a given level of risk**.

This idea is based on **Markowitz Mean-Variance Optimization**, which is a trade-of of:

- Expected return: $\mathbb{E}[R_p]$
- Portfolio volatility: $\sigma_p = \sqrt{\mathbf{w}^T \Sigma \mathbf{w}}$

The result is a **efficient frontier**, where every point is an optimized portfolio. The point with the **lowest volatility** is called **minimum-variance-portfolio**. All portfolios with a lower return than that are **not efficient**

Tangency Portfolio & Sharpe Ratio

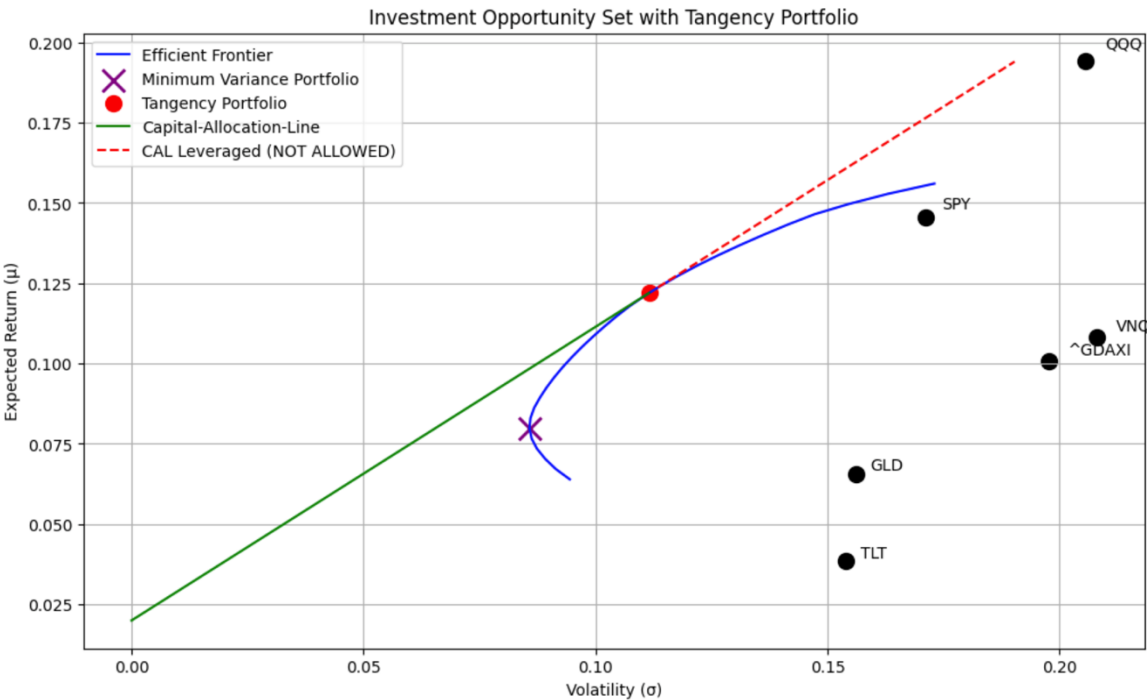
Among all efficient portfolios, the **Tangency Portfolio** is special: it maximizes the **Sharpe Ratio**, which measures risk-adjusted performance:

$$\text{Sharpe Ratio} = \frac{\mathbb{E}[R_p] - R_f}{\sigma_p}$$

It represents the **point of tangency** between the Efficient Frontier and the **Capital Allocation Line (CAL)**, which starts at the risk-free rate and shows all combinations of risk-free and risky investments.

This visualization captures the essence of **modern portfolio theory**:

By selecting portfolios along the CAL, investors can customize exposure based on their risk tolerance, while always staying within the **most efficient region** of the opportunity set.



```
# 3. Efficient Frontier Calculation
def calculate_efficient_frontier(mu, Sigma):
    target_returns = np.linspace(mu.min(), mu.max(), 50)
    efficient_weights = []

    for target in target_returns:
        constraints = [
            {'type': 'eq', 'fun': lambda w: np.sum(w) - 1},
            {'type': 'eq', 'fun': lambda w: portfolio_return(w) - target}
        ]

        result = minimize(
            portfolio_volatility,
            x0=np.ones(len(mu))/len(mu),
            bounds=[(0,0.4) for _ in range(len(mu))], # dont invest more than 50% into one asset
            constraints=constraints
        )

        if result.success:
            efficient_weights.append(result.x)

    return efficient_weights

efficient_weights = calculate_efficient_frontier(mu, Sigma)
efficient_returns = [portfolio_return(w).item() for w in efficient_weights]
efficient_volatilities = [portfolio_volatility(w).item() for w in efficient_weights]

# 4. Tangency Portfolio
def find_tangency_portfolio(mu, Sigma, risk_free_rate):
    constraints = [{'type': 'eq', 'fun': lambda w: np.sum(w) - 1}]
    bounds = [(0,0.4) for _ in range(len(mu))]

    result = minimize(
        lambda w: -portfolio_sharpe(w),
        x0=np.ones(len(mu))/len(mu),
        bounds=bounds,
        constraints=constraints
    )
```

Tangency Portfolio Composition:

GLD: 14.31%
QQQ: 40.00%
SPY: 14.47%
TLT: 27.98%
VNQ: 0.00%
^GDAXI: 3.23%

Tangency Portfolio Performance:

Expected Return: 12.21% p.a.
Volatility: 11.17% p.a.
Sharpe Ratio: 0.91

Optimal Complete Portfolio (Risk Aversion = 9):

Allocation to Tangency Portfolio: 90.95%
Allocation to Risk-Free Asset: 9.05%
Expected Return: 11.29% p.a.
Volatility: 10.16% p.a.

3) Tangency Portfolio vs. Optimal Complete Portfolio

The **Tangency Portfolio** is the portfolio on the efficient frontier that maximizes the **Sharpe Ratio**:

$$\text{Sharpe Ratio} = \frac{\mathbb{E}[R_p] - R_f}{\sigma_p}$$

It represents the best trade-off between risk and return across risky assets. Visually, it is the point where the **Capital Allocation Line (CAL)** is a tangent to the **efficient frontier**.

Why are the assets distributed this way?

The optimal risky portfolio allocation (Tangency Portfolio) is the result of **mean-variance optimization** under constraints (in our case max 40% per asset, no short-selling). The current composition is:

- **QQQ (Nasdaq 100)**: 40.0%
- **TLT (Long-term U.S. Bonds)**: 28.0%
- **SPY (S&P 500)**: 14.5%
- **GLD (Gold)**: 14.3%
- **^GDAXI (German Index)**: 3.2%
- **VNQ (REITs)**: 0.0%

Interpretation:

- **QQQ** reaches the **maximum cap of 40%**, showing that the optimizer would put even more weight based on its return and Sharpe ratio—but is **capped to limit concentration risk**.
- This constraint is **important** to avoid overexposure to a single sector or market (in our case U.S. tech), which could make the portfolio super sensitive to sector-specific shocks.
- **TLT** complements equities as a hedge (negatively correlated), reducing portfolio volatility.
- **VNQ** and **^GDAXI** play minor roles due to either correlation redundancy or suboptimal risk-return profiles.
- **GLD** is included as a non-correlated diversifier.

What is the Optimal Complete Portfolio?

The **Optimal Complete Portfolio** combines the tangency portfolio with the **risk-free asset**, it depends on the investor's **risk aversion (A)**.

The optimal share invested in the tangency portfolio is calculated as:

$$y^* = \frac{\mathbb{E}[R_p] - R_f}{A \cdot \sigma_p^2}$$

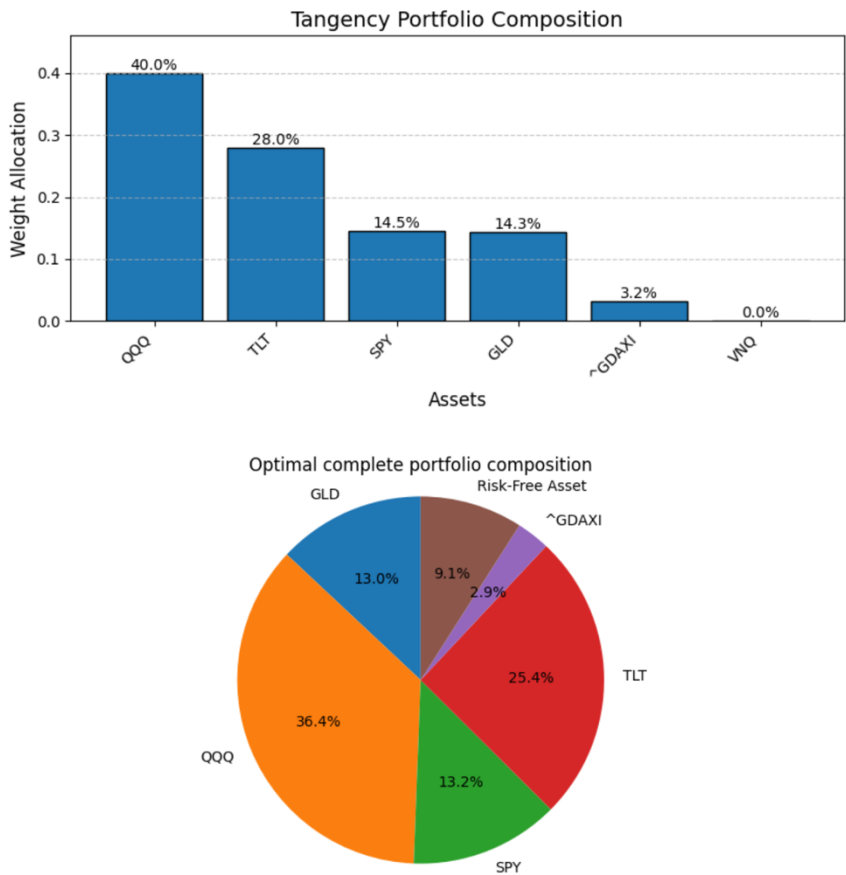
Given a (high) risk aversion of $A = 9$, the optimal composition is:

- **90.95%** Tangency Portfolio
- **9.05%** Risk-Free Asset

This portfolio lies on the **Capital Allocation Line (CAL)** and delivers a personalized mix of return and risk. We are very risk-averse.

Summary & Interpretation

- The **Tangency Portfolio** provides the most efficient mix of risky assets.
- The **Optimal Complete Portfolio** adapts this mix considering our personal risk tolerance.
- In this case, the **expected return is 11.29%** with **volatility of 10.16%**, resulting in a **Sharpe Ratio of 0.91**.
- The pie chart clearly shows a **concentrated exposure to tech (QQQ)** and **bonds (TLT)**, while still maintaining diversification through **gold** and **international equities**.
- **Capping asset weights** improves robustness by avoiding excessive focus on a single asset (Nasdaq100ETF).



4) Risk Decomposition Using Linear Regression

To understand the **source of risk and return** in a portfolio, we decompose its performance into **systematic (market-related)** and **idiosyncratic** components.

This is done using **linear regression**, where the portfolio's excess return is regressed on the market's excess return:

$$R_p - R_f = \alpha + \beta(R_m - R_f) + \epsilon$$

- R_p : Portfolio return
- R_f : Risk-free rate
- R_m : Market return
- α : Idiosyncratic (non-market) return
- β : Sensitivity of the portfolio to the market (systematic risk)
- ϵ : Residual (unexplained) return

We can also decompose the **portfolio's total risk** (volatility) as:

$$\sigma_p^2 = (\beta \cdot \sigma_m)^2 + \sigma_\epsilon^2$$

- σ_p : Total portfolio standard deviation
- σ_m : Market standard deviation
- σ_ϵ : Idiosyncratic risk (residual volatility)

This allows us to quantify how much of the portfolio's behavior is **explained by the market**, and how much is **special to the portfolio strategy**.

```
# Run OLS
X = sm.add_constant(market_excess)
model = sm.OLS(portfolio_excess, X).fit()

print(model.summary())

# Coefficients
beta = model.params.iloc[1]
alpha = model.params.iloc[0]

# Calculate Return Composition
systematic_return = beta * market_excess.mean() * 252 + risk_free_rate
idiosyncratic_return = alpha * 252
total_return = systematic_return + idiosyncratic_return
```

OLS Regression Results						
Dep. Variable:	y	R-squared:	0.680			
Model:	OLS	Adj. R-squared:	0.680			
Method:	Least Squares	F-statistic:	7866.			
Date:	Sat, 24 May 2025	Prob (F-statistic):	0.00			
Time:	11:22:32	Log-Likelihood:	15226.			
No. Observations:	3707	AIC:	-3.045e+04			
Df Residuals:	3705	BIC:	-3.043e+04			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0001	6.55e-05	2.208	0.027	1.62e-05	0.000
SPY	0.5401	0.006	88.691	0.000	0.528	0.552

	Value
Total Return	0.122125
Systematic Return (β)	0.085688
Idiosyncratic Return (α)	0.036436
Total Risk	0.111698
Systematic Risk ($\beta\sigma_m$)	0.092095
Idiosyncratic Risk (σ_ϵ)	0.063205
Beta (β)	0.540128
Alpha (α)	0.036436
R-squared	0.679805

OLS Regression Output:

- **Beta (β) = 0.540**
→ The portfolio reacts moderately to market movements. A 1% change in the market leads to a 0.54% change in the portfolio.
- **Alpha (α) = 0.0364**
→ The portfolio generates a **positive idiosyncratic return** of 3.64% that is not explained by market risk.
- **R² = 0.680**
→ 68% of the portfolio's excess return variability is explained by market movements, while 32% stems from specific portfolio characteristics (non-systematic risk).

Significance of Coefficients:

- Both **β** and **α** are **statistically significant** (p-values < 0.05).
→ This confirms that the regression model captures both systematic and idiosyncratic return components.

5) Expected vs. Realized Performance of the Tangency Portfolio

Now we compare the **expected performance** (on the training sample) with the **realized performance** (on the test sample).

Method:

- Train/Test split: 70 % / 30 %

Expected Return: 0.1376, Expected Volatility: 0.1000
Realized Return: 0.0861, Realized Volatility: 0.1352

Interpretation of Out-of-Sample Results

- **Expected Return:** 13.76% p.a. vs. **Realized Return:** 8.61% p.a.
- **Expected Volatility:** 10.00% p.a. vs. **Realized Volatility:** 13.52% p.a.

The realized performance is significantly different from our expectations:

- The return is **lower than expected**, indicating the model may have overestimated future profitability.
- The realized risk is **higher**, suggesting stronger market fluctuations or underestimation of variance.

This shows the importance of **out-of-sample testing**:

- Even if in-sample performance looks strong, **future outcomes are uncertain**.
- Differences can result from **changing market conditions**, **parameter estimation errors**, or **overfitting** to old data.